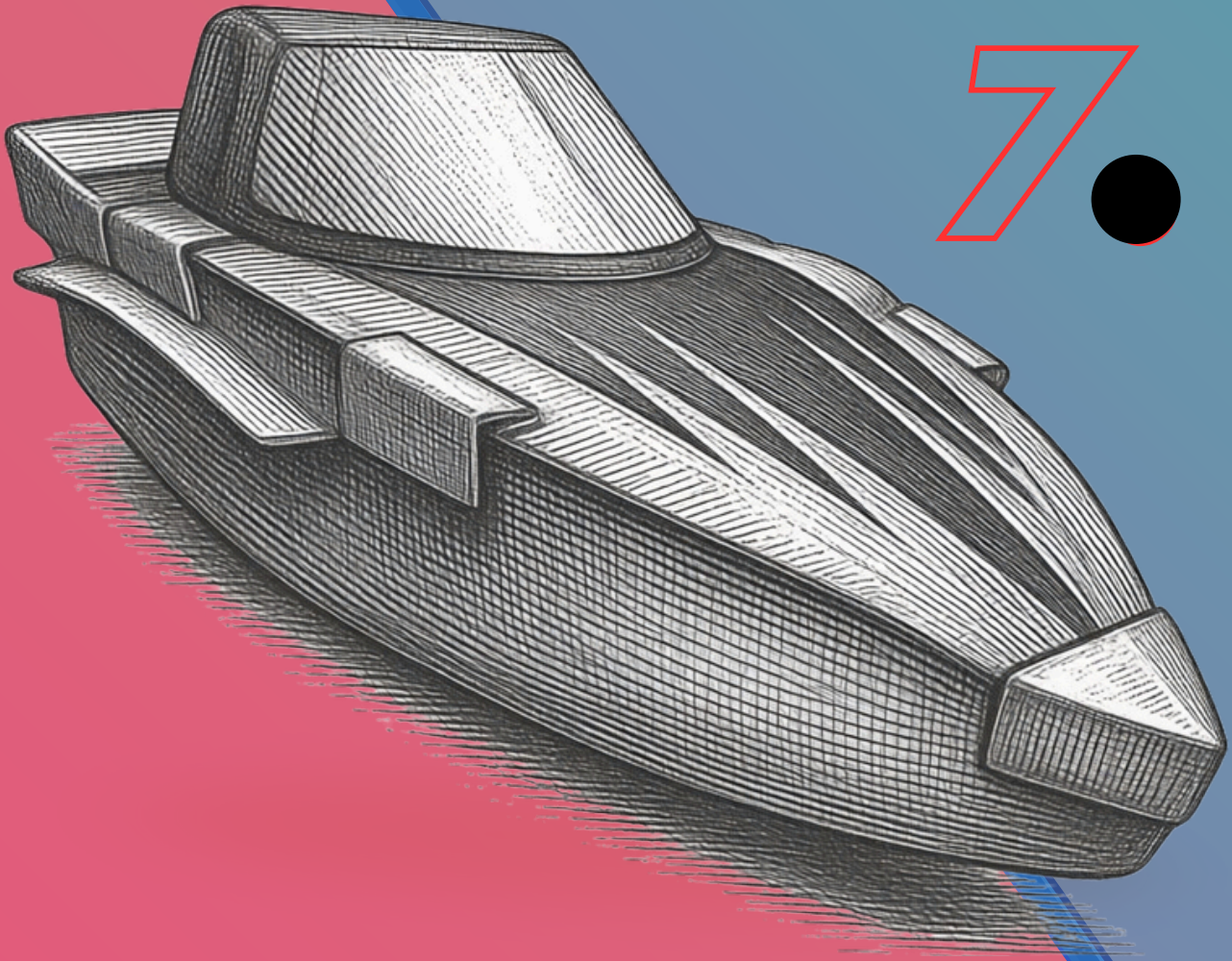




FLAREON

7.1



TEMPEST 2026 and Wavez 2026 (IITM)

Narrow v-hull integrated with an fixed propulsion system for superior maneuverability. It's controlled via a 6-channel RC system with PID tuning, enabling precision thrust vectoring and agile navigation



IN SEARCH OF INNOVATION

BUILDNEO: A student-led research and development initiative at the Indian Maritime University, Visakhapatnam, focused on innovation and practical skill development in naval architecture. Founded by ambitious alumni, it thrives on a network of past and present students collaborating on working prototypes and research. The initiative features a dynamic leadership team, including roles in external communications, electronics, technical documentation, coding, CAD modeling, fabrication, additive manufacturing, and social media. By promoting project flexibility and creativity, BuildNEO prepares members for academic and industrial challenges, aiming to become a hub of talent and innovation in the maritime industry.

FLAREON 7.1

This project involves the development of a compact, remotely operated surface vessel designed for short-range operations in controlled water environments. It is powered by an efficient brushless DC motor that ensures reliable propulsion with low power consumption, and it is wirelessly controlled using a radio communication module for accurate navigation. The vessel features a streamlined hull for improved stability, smooth movement, and hydrodynamic efficiency. Its design ensures effective power management and consistent performance, while also allowing modular expansion for adding sensors or other equipment. This system is suitable for environmental monitoring, aquatic research, and small-scale surface inspection tasks.

OBJECTIVE

The objective of this project is to design and develop a compact, efficient, remotely operated watercraft capable of performing short-range surface tasks with stable operation in calm water conditions. It also provides a base for future upgrades such as automation, data collection, and sensor-based navigation.

SPECIFICATIONS

- 50cm (Length) × 21 cm (Width) × 15 cm (Depth)
- Draft - 4cm
- A2212 1400KV BLDC motor
- SimonK 30A ESC
- 11.1V 2200mAh 60C LiPo battery
- FlySky FS-iA6B 2.4GHz 6CH receiver
- Max speed - 7.1 knots
- Range: 600 m
- Endurance: 30–70 minutes

FABRICATION TEMPLATE

The fabrication process begins with the preparation of a paper template representing the boat's principal dimensions, including length, breadth, and depth, to enable proper visualization of the design. The template is then refined by carefully verifying proportions and making necessary adjustments to ensure accuracy, balance, and structural suitability. Once finalized, the template is transferred onto a foam board by precisely tracing the outlines of all components. The foam board is cut along the traced lines using suitable tools, ensuring safety and clean edges. The cut components are assembled using marine adhesive to form a strong and water-resistant structure. Finally, electronic systems are installed, and a trial test is conducted to evaluate stability, balance, and overall performance.

